TEXORA · COURSE NOTES · LECTURE SHEET

Applications of Differential Geometry

The geometry of curves measured with calculus — arc length, curvature, the radius and centre of curvature, evolutes, and torsion for curves in space.

MATH-1101 · MATHEMATICS-I · TOPIC 3 · LEVEL 1 TERM 1

1. Arc length

Differential geometry asks calculus questions about curves: how long is this piece of curve, how sharply does it bend here, and where is the centre of the circle that best fits it? The natural measure along a curve is the **arc length** s, measured from some fixed point of the curve. For a plane curve y = f(x),

$$s = \int \sqrt{1 + (dy/dx)^2}\, dx, \quad \text{so that} \quad ds/dx = \sqrt{1 + (dy/dx)^2}.$$

For a curve given parametrically as x = x(t), y = y(t), the integrand becomes $\sqrt{(dx/dt)^2 + (dy/dt)^2}\, dt$; for polar curves r = r(θ), it becomes $\sqrt{r^2 + (dr/d\theta)^2}\, d\theta$. Measuring position by arc length rather than by x or t gives the *intrinsic* description of a curve, independent of how it happens to be drawn on axes.

WORKED EXAMPLE — ARC LENGTH

Find the length of $y = (2/3)\, x^{3/2}$ from x = 0 to x = 3. Here dy/dx = √x, so ds/dx = √(1 + x) and $s = \int_0^3 \sqrt{1 + x}\, dx = (2/3)\, [(1 + x)^{3/2}]_0^3 = (2/3)\,(8 - 1) =$ **14/3**.

2. Curvature

Curvature measures how fast a curve turns. If ψ is the angle the tangent makes with the x-axis, the curvature at a point is the rate of turning per unit arc,

$$\kappa = |d\psi/ds|, \quad \text{and for } y = f(x): \quad \kappa = |y''| / (1 + y'^2)^{3/2}.$$

A straight line has κ = 0 everywhere (it never turns). A circle of radius a has constant curvature κ = 1/a — small circles bend sharply, large ones gently, which is why curvature is compared against circles. The parametric form is $\kappa = |x'y'' - y'x''| / (x'^2 + y'^2)^{3/2}$.

2.1 Radius and centre of curvature

The **radius of curvature** $\rho = 1/\kappa$ is the radius of the **circle of curvature** (osculating circle) — the circle that touches the curve at the point and matches its bending there. Its centre, the **centre of curvature**, lies on the normal to the curve at distance ρ from the point, on the concave side. For $y = f(x)$ the centre $(\bar{x}, \bar{y})$ is

$$\bar{x} = x - y'(1 + y'^2)/y'', \quad \bar{y} = y + (1 + y'^2)/y''.$$

> **WORKED EXAMPLE — CURVATURE OF A PARABOLA**
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> For $y = x^2$: $y' = 2x$ and $y'' = 2$, so $\kappa = 2 / (1 + 4x^2)^{3/2}$. At the vertex ($x = 0$) the curvature is greatest: $\kappa = 2$, hence $\rho = 1/2$, and the centre of curvature is $(0, 1/2)$ — the circle of radius $1/2$ sitting on the vertex and matching the parabola's bend there. Far from the vertex the parabola straightens and κ falls toward 0.

3. Evolutes and involutes

As one travels along a curve, the centre of curvature moves too, and its path is a new curve called the **evolute** of the original. Running the relationship backwards, the original curve is an **involute** of its evolute: an involute is traced by the end of a taut string being unwound from the evolute. Involutes are not a curiosity — the profiles of most gear teeth are involutes of a circle, because that shape makes mating gears roll together with constant velocity ratio. The evolute of a parabola is a semicubical parabola; the evolute of a circle is a single point, its centre.

4. Curves in space: the Frenet frame and torsion

A space curve needs two measures instead of one. At each point, three mutually perpendicular unit vectors form the **Frenet frame**: the **tangent** T (direction of motion), the **principal normal** N (direction in which the tangent is turning), and the **binormal** $B = T \times N$. **Curvature** $\kappa = |dT/ds|$ still measures bending within the curve's instantaneous plane, while **torsion** τ measures how the curve twists out of that plane — a plane curve has $\tau = 0$ identically, and a helix has both κ and τ constant. The rates of change of the frame are summarised by the **Serret–Frenet formulas**:

$$dT/ds = \kappa N, \quad dN/ds = \tau B - \kappa T, \quad dB/ds = -\tau N.$$

Together, κ and τ as functions of arc length determine a space curve completely (up to its position in space) — the fundamental theorem of curve theory, and the three-dimensional completion of this topic.

5. Why an engineer cares

Curvature is the mathematics of bending, and bending is everywhere in engineering: the profile of a cam or gear tooth, the path of a road or railway through a curve (where curvature sets safe speed), the bending of yarns and fibres around guides and needles in textile machinery, and the drape of a fabric, which is governed by how its yarns curve. The formulas of this topic are the tools for putting numbers on “how sharply does it bend here?”

6. Key points and summary

- Arc length accumulates as $ds = \sqrt{(1 + y'^2)}\,dx$ (with parametric and polar variants), and position along a curve is measured intrinsically by s.
- Curvature $\kappa = |y''| / (1 + y'^2)^{3/2}$ measures turning per unit arc; $\rho = 1/\kappa$ is the radius of the osculating circle, whose centre is the centre of curvature.
- The locus of centres of curvature is the evolute; unwinding a string from a curve traces its involute — the shape of gear teeth.
- Space curves need the Frenet frame and a second invariant, torsion τ, measuring twist out of the osculating plane; κ and τ together determine the curve.

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